

Operational semantics for Prolog with Cut in Rocq and its application to determinacy analysis

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Abstract

We mechanize two operational semantics for PROLOG with cut and prove them equivalent, in the Rocq prover. The first semantics closely models the ELPI’s implementation, it is based on a stack of choice points and is well suited for studying optimizations employed by PROLOG abstract machines. The second semantics is instead based on and-or trees, in which the branches pruned by the cut operator are made explicit.

We use the tree-based semantics (1) to prove properties about the effect of the cut operator in the evaluation of choice points, since the rules from classical logic do not apply, and (2) to reason about the determinacy analysis introduced in Elpi version 3.0, which statically identifies computations that produce at most one result.

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1 Introduction

ELPI is a dialect of λ PROLOG (see [14, 15, 7, 12]) used as an extension language for the ROCQ prover (formerly the COQ proof assistant). ELPI has become an important infrastructure component: several projects and libraries depend on it [13, 3, 4, 19, 8, 9]. Examples include the Hierarchy-Builder library-structuring tool [5] and Derive [17, 18, 11], a program-and-proof synthesis framework with industrial applications at SkyLabs AI.

Starting with version 3, ELPI is equipped with a static analysis for determinacy [10] to help users tame backtracking. ROCQ users are familiar with functional programming but not necessarily with logic programming, in particular uncontrolled backtracking is a common source of inefficiency and makes debugging harder. The determinacy checker identifies predicates that behave like functions, i.e., predicates that commit to their first solution and leave no *choice points* (places where backtracking could resume).

This paper reports our first steps towards a mechanization, in the ROCQ prover, of the determinacy analysis from [10]. We focus on the control operator *cut*, which is useful to restrict backtracking but makes the semantics depart from a pure logical reading.

We formalize two operational semantics for PROLOG with cut. The first is a stack-based semantics that closely models ELPI’s implementation and is similar to the semantics mechanized by Pusch in ISABELLE/HOL [16] and to the model of Debray and Mishra [6, Sec. 4.3]. This stack-based semantics is a good starting point to study further optimizations used by standard PROLOG abstract machines [20, 1], but it makes reasoning about the scope of cut difficult. To address this limitation we introduce a tree-based semantics in which the branches pruned by *cut* are explicit and we prove the two semantics equivalent. Using the



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```

Inductive P := IP of nat. Inductive D := ID of nat. Inductive V := IV of nat.

Inductive Tm :=
  | Tm_P of P
  | Tm_D of D
  | Tm_V of V
  | Tm_App of Tm & Tm.

Record Unif := {
  unify : Tm → Tm → Σ → option Σ;
  matching : Tm → Tm → Σ → option Σ;
}.

```

■ Figure 1 Tm types

■ Figure 2 Unif type

tree-based semantics, we then prove some properties about the impact of evaluating the *cut* in query disjunctions. Unlike in classical logic, $q_1 \vee q_2$ is not equivalent to $q_2 \vee q_1$, since q_2 may be cut-away by q_1 . Moreover, we show that if every rule of a predicate passes the determinacy analysis, then a call to that predicate does not leave any choice points.

2 Preliminaries: syntax and informal semantics

Before going to the two semantics, we show the piece of data structure that are shared by the them. The smallest unit of code that we can use in the language is an atom. The atom inductive (see Definition 1) is either a cut or a call. A call carries a term (see Figure 1). A term (Tm) is either a predicate, a datum, a variable or the binary application of two terms.

```

Inductive A := cut | call : Tm → A. (1)
Record R := mkR { head : Tm; premises : list A }. (2)
Record P := { rules : seq R; sig : sigT }. (3)
Definition Σ := {fmap V → Tm}. (4)
Definition bc : P →  $\mathcal{F}_V$  → Tm → Σ →  $\mathcal{F}_V$  * seq (Σ * seq A) := (5)

```

A rule (see Definition 2) has a head of type term and a list of premises, that are a list of atoms. A program (see Definition 3) is made by its rules and the signatures of predicates, **sigT** is a mapping from predicates to their signatures, signatures represent the definition of type from the simply typed lambda calculus, i.e. it is either a base type or an arrow for term application. We decorate arrows to know the mode (input or output) of the application argument.

A substitution (see Definition 4) is a mapping from variables to terms. It is the output of a successful query. The empty substitution is noted ε .

The backchain function (bc, see Definition 5) takes: a program P ; a set fv of variables ($\mathcal{F}_V$); a query q ; and the substitution σ . A backchain is performed when the current atom is a **call**. In this case, it start by recursively *dereferencing* the variables in σ , i.e. replacing all the variables in the query by their term mapping in σ .

TODO: free variables and program freshness

Then it takes the head of the term, which should be rigid, i.e. a predicate p , and look for the signature s of p in the program. Then, for each rule r in P , it unifies the head of r with q . Unification is possible thanks to the unificator U , whose type is shown in Figure 2. U explains how to unify terms up to standard unification (for output terms) or matching (for input terms). The function *unif* (resp. *matching*) unifies (resp. matches) the two terms under σ , it return an extension of σ if the two terms can be unified (resp. matched), and None otherwise. The result of a backchain is made by filtering the rules in P that can be used on the given query. In particular, for each rule filtered rules r , it returns a pair containing the extension of σ making the head of r and q unify and the list of premises of r .

```

f 1 2.    f 2 3.
r 0 0.    r 0 1.    r 0 2 :- !.
g A C :- r A B, f B C, !.    % r1
g D F :- f D E, f E F.      % r2

```

■ **Figure 3** Small ELPI program example

```

Inductive tree :=
| KO | OK | TA of A
| Or of option tree & Σ & tree
| And of tree & seq A & tree.

```

■ **Figure 4** Tree inductive

2.1 The cut operator

ELPI is a dialect of λ PROLOG with the hard cut operator and, in the language we have shown before, *cut* is an atom. The semantics of the cut operator adopted in the ELPI language corresponds to the *hard cut* operator of standard SWI-PROLOG. This operator has two primary purposes. First, it eliminates all alternatives that are created either simultaneously with, or after, the introduction of the cut into the execution state.

The ELPI program we will use as an example in the following sections is shown in Figure 3.

To illustrate the high-level description of *cut*, consider the program P shown in Figure 3 and the query $q = g \ 0 \ R$. Both rules for g can be used on the query q . Backchaining q in P from the empty (ε) substitution returns the list $[(\sigma_1, [r \ A \ B, f \ B \ C, !]), (\sigma_2, [f \ D \ E, f \ E \ F])]$ with $\sigma_1 := \{A \mapsto 0, B \mapsto C\}$ and $\sigma_2 := \{D \mapsto 0, F \mapsto R\}$. The two elements of the list are generated respectively from the rules r_1 and r_2 . For simplicity, we have taken a program where the variable of each pair rule are disjoint and a query with variables disjoint from the program, this avoid the *bc* function to fresh the program variable.

The next step of the interpretation corresponds to the backchaining of $r \ X \ Y$ from σ_1 . This returns $[(\sigma_3, []), (\sigma_4, []), (\sigma_5, [!])]$ with $\sigma_3 := \sigma_1 + \{B \mapsto 0\}$, $\sigma_4 := \sigma_1 + \{B \mapsto 1\}$ and $\sigma_5 := \sigma_1 + \{B \mapsto 2\}$. The three lists of premises are empty since all rules for r have no premises.

Backchaining $f \ B \ C$ in σ_3 gives no solution, therefore, by backtracking, we backchain $f \ B \ C$ in σ_4 . This gives a solution, namely $\sigma_6 := \sigma_4 + \{R \mapsto 2\}$.

We now reach the cut operator and can see its double side behavior. In the current execution state, we have still two unexplored alternatives: we can backtrack on the third solution for r that assign B to 2 and on the rules r_2 . We can note, however, the first alternative have been generated after the cut introduction, and the second alternative have been generated at the same moment of the cut. Therefore both are cut away leaving no choice point. The final complete solution is: $\{A \mapsto 0, B \mapsto 1, C \mapsto R, R \mapsto 2\}$, where A , B and C can be ignored since they are the result internal unification.

We propose two formal, operational semantics for a logic program with cut. The two semantics are based on different syntaxes, the first syntax (called *tree*) exploits a tree-like structure and is ideal both to have a graphical view of its evolution while the tree is being interpreted and to prove lemmas over it. The second syntax, called *elpi*, is the ELPI's syntax and has the advantage of reducing the computational cost of cutting and backtracking alternatives by using shared pointers. We aim to prove the equivalence of the two semantics together with some interesting lemmas of the cut behavior.

Notations used in the paper In the following section we note $\mathbb{B}$ for the boolean type, and $\top$ and $\perp$ for **true** and **false**. The option instances are noted $\bullet t$ for *Some t* and $\square$ for *None*.

spiegare nota-
tion for list

```

118 Fixpoint get_end s A :  $\Sigma$  * tree :=
119   match A with
120   | TA _ | KO | OK  $\Rightarrow$  (s, A)
121   | Or  $\square$   $\sigma_1$  B  $\Rightarrow$  get_end  $\sigma_1$  B
122   | Or  $\bullet$  A _  $\Rightarrow$  get_end s A
123   | And A _ B  $\Rightarrow$ 
124     let (s', pA) := get_end s A in
125     if pA == OK then get_end s' B
126     else (s', pA)
127   end.

128 Definition get_subst s A := (get_end s A).1.
129 Definition path_end A := (get_end  $\varepsilon$  A).2.
130 Definition success A := path_end A == OK.
131 Definition failed A := path_end A == KO.
132 Definition path_atom A :=
133   if path_end A is TA _ then  $\top$ 
134   else  $\perp$ .

```

(a) Definition of *get_end*(b) Functions from *get_end*■ **Figure 5** *get_end* and derives functions

117 3 And-Or Tree semantics

118 *The tree state* The tree inductive, shown in Figure 4. We distinguish 5 main cases: *KO*, *OK*,
 119 and are special meta-symbols representing, respectively, the basic failed and a successful
 120 trees. These symbols are considered meta because they are internal intermediate symbols
 121 used to give structure to the tree.

122 The *TA* constructor (acronym for tree-atom) is the constructor of atoms in the tree. They
 123 are either the cut operator or a call, in these cases, the interpreter should evaluate the atom
 124 by either cutting the subtrees in the scope of the cut or backchaining the call in the program.

125 The two recursive cases of a tree are the *Or* and *And* nodes. The *Or* node $A \vee B_\sigma$ denotes
 126 a disjunction between two trees *A* and *B*. The first branch is optional, if absent it represents
 127 a dead tree, i.e. a tree that has been entirely explored. The second branch is annotated
 128 with a suspended substitution σ so that, upon backtracking to *B*, σ is used as the initial
 129 substitution for the execution of *B*.

130 The *And* node $A \wedge_{B_0} B$ represents a conjunction of two trees *A* and *B*. We call B_0 the
 131 reset point for *B*; it is used to restore the state of *B* to its initial form if a backtracking
 132 operation occurs on *A*. An example of this behavior is shown in Section 3.2

133 A graphical representation of a tree is shown in Figure 6a. To make the graph more
 134 compact, the *And* and *Or* nodes are n-ary rather than binary, and the children of these
 135 nodes associate to the right. The *KO* node acts as the neutral elements in the *Or* list, while
 136 *OK* is the neutral element of the *And* list.

137 *The tree interpreter* The interpretation of a tree is performed by two main routines: *step*
 138 and *next_alt* that traverse the tree depth-first, left-to-right. Then, then *runT* inductive
 139 makes the transitive closure of *step* and *next_alt*: it iterates the calls to its these
 140 auxiliary functions as much as needed. In Definitions 7–9 we give the types contrats of these
 141 symbols where $\mathcal{F}_v$ is a set of variable names.

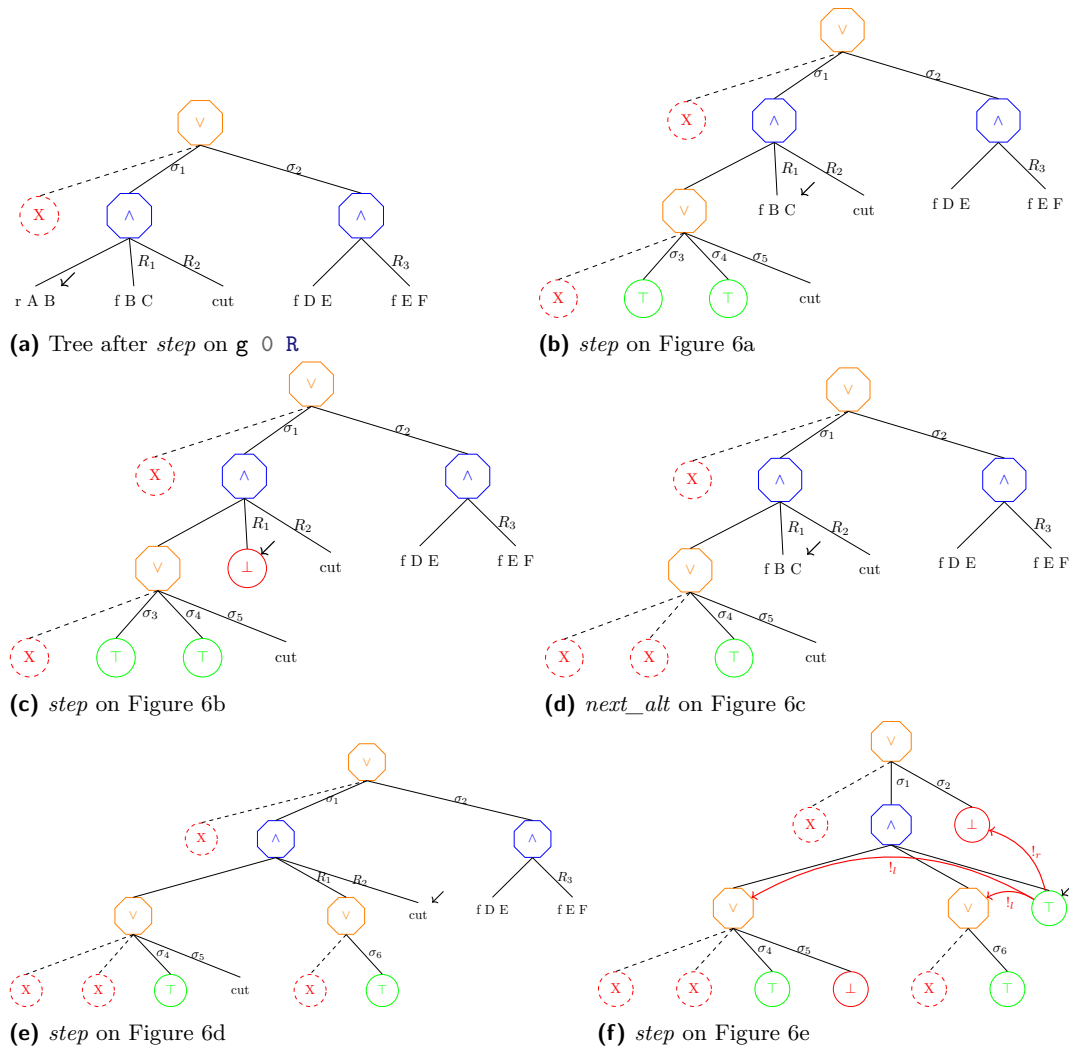
142 **Inductive** tag := Expanded | CutBrothers | Failed | Success. (6)

143 **Definition** step : $\mathbb{P} \rightarrow \mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree} \rightarrow (\mathcal{F}_v * \text{tag} * \text{tree}) :=$ (7)

144 **Definition** next_alt : $\mathbb{B} \rightarrow \text{tree} \rightarrow \text{option tree} :=$ (8)

145 **Inductive** runT (p : $\mathbb{P}$): $\mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree}$
 $\rightarrow \text{option } (\Sigma * \text{option tree}) \rightarrow \mathbb{B} \rightarrow \mathcal{F}_v \rightarrow \text{Prop} :=$ (9)

146 The tree interpreter, as in prolog, explores the tree in DFS strategy, to discover the
 147 substitution and the leaf of the tree that should be interpreted. These two pieces of
 148 information can be retrieved for a substitution *s* and a tree *A* thanks to the *get_end* routine,



■ **Figure 6** Some tree representations¹

149 shown in Figure 5a. If the tree is already a leaf, *get_end* returns its inputs. Otherwise, if the
 150 tree is a disjunction, the path continues on the left subtree, if it exists, otherwise *get_end*
 151 continues on the rhs using the substitution stored in the *Or* node. In the case of a conjunction,
 152 if *get_end* of the lhs returns the tree *OK*, then the *get_end* continues in the rhs, otherwise
 153 we return the result of the lhs. We derive some useful definition from *get_end*. They are
 154 listed in Figure 5b.

155 We have pointed by a black arrow the *path_end* of all the trees in Figure 6.

fare fresh rules
in Figure 6

156 3.1 The *step* procedure

157 The *step* procedure takes as input a program, a set of variables, a substitution, and a tree,
 158 and returns an updated set of variables, a *tag*, and an updated tree.

¹ The substitution $\sigma_1 \dots \sigma_6$ are from Section 2.1, R_1, R_2 and R_3 are the reset points and correspond respectively to the list of atoms $[f \ Y \ Z, !], [!]$ and $[f \ Y \ Z]$

The set of variables v given in input to `step` are used and updated when a backchaining operation takes place. A call to `step` produces a good result if v contains all the variables in the tree. This guarantees that any call to `bc` returns a list of atoms whose variables are disjoint from the variables of the current working tree.

The *tag* (see Definition 6) indicates the kind of an internal tree step: **CutBrothers** denotes the interpretation of a superficial cut, i.e., a cut whose parent nodes are all *And*-nodes. **Expanded** denotes the interpretation of non-superficial cuts or predicate calls. **Failure** and **Success** are returned for, respectively, *failed* and *success* trees.

The step procedure is intended to evaluate atoms, that is, it leaves unchanged the tree iff its *path_end* is not an atom, i.e. if it is a success- or a failed-tree.

Lemma succ_step_iff: $\forall p \ v \ s \ A, \text{success } A \leftrightarrow \text{step } p \ v \ s \ A = (v, \text{Success}, A).$ (10)

Lemma fail_step_iff: $\forall p \ v \ s \ A, \text{failed } A \leftrightarrow \text{step } p \ v \ s \ A = (v, \text{Failed}, A).$ (11)

The interpretation of a call to a term c starts by calling the `bc` function on c . The result of the backchain is a list l . If l is empty then the current call is replaced by the *KO* node, otherwise it is replaced by a right-skewed tree made by *Or* inner-nodes and each leave correspond to a list of atom e in l . If e is empty then it is mapped to the *OK* tree, otherwise it is mapped to a right-skewed tree made by *And* inner-nodes.

In Figure 6a, we provide a clear example of a call to `step` on the query $q \ 0 \ R$ on the program P in Figure 3. Let c_0 , c_1 and c_2 be the children of the root. c_0 , by construction is the *None* tree, while c_1 and c_2 correspond respectively to the premises of rules r_1 and r_2 in P . We need the first *None* subtree so that we can attach a substitution to c_1 . In particular, we note that the c_1 (resp. c_2) are related to the substitution σ_1 (resp. σ_2). The value of σ_1 and σ_2 are the same as in Section 2.1. The reset points in the tree are marked with the R symbol. $R_1 := [f \ Y \ Z, !]$, $R_2 := [!]$ and $R_3 := f \ Y \ Z$, they are essentially the list of atoms that allowed to generate the current subtree. Other example of call to the `step` procedure triggering a backchain are Figures 6b, 6c, and 6e.

The interpretation of a cut has three main impacts: at first the cut is replaced by the *OK* node, then some special subtrees, in the scope of the *Cut*, are cut away: in particular we need to soft-kill the left-siblings of the *Cut* and hard-kill the right-uncles of the *Cut*.

► **Definition 1** (Left-siblings (resp. right-sibling) and Right-uncles). *Given a node A , the left-siblings (resp. right-sibling) of A are the list of subtrees sharing the same parent of A and that appear on its left (resp. right). The right-uncles of A are the list of right-sibling of the father of A .*

► **Definition 2** (Hard-kill, $!_r$ and Soft-kill, $!_l$). *Given a tree t , hard-kill replaces the given subtree with the *KO* node. If t is a success-tree, then soft-kill replaces with *KO* all subtrees that are not part of the path in t leading to the *OK* node.*

An example of the impact of the cut is show in Figure 6f. In Figure 6e, we see that the *path_end* is a cut. The cut is replaced by the *OK* node and it need to cut-away some alternatives in the tree. The red arrows indicate the scope of the cut, they are labeled with $!_r$ and $!_l$ to explain which kind of cut has been performed on each pointed subtree. We note that, after the evaluation of a cut, the path leading to the cut node is not affected by $!_l$, the success under the substitution σ_4 is kept, while the success under the substitution σ_5 is replaced by the *KO* node. If we refer to the paragraph in Section 2.1, we are discarding the third rule of the r predicate.

3.2 The *next_alt* procedure

The *step* procedure alone is not sufficient to reproduce entirely the behavior of the full expected prolog interpreter. As proved in Definition 10 and Definition 11, *step* does not make any progression on the tree being interpreted. To solve this problem, we need a routine capable of backtracking in case of failures. This routine return the backtracked tree if it exists, *None* otherwise.

In Figure 6c, we are facing a failure: the interpreter has evaluated the atom $\mathbf{f} \ X \ Y$ under the substitution σ_3 which assigns Y to 0, but, in Figure 3, we have no rule satisfying this query. The role of *next_alt*, in this example, is to kill the path leading to this failure, while resetting the failure to its original shape. We see how the reset point plays a major role in this case. At first we are killing the *OK* node under the substitution σ_3 , since this tree have been entirely explored and led to no solution, then we replace the *KO* node with the information stored in R_1 . That is, thanks to R_1 , we are restoring the $\mathbf{f} \ Y \ Z$ call, so that we are able to evaluate this node under the substitution σ_4 . More concretely, we have uncessesfully tried rule $\mathbf{r} \ 0 \ 0$, therefore, we need to continue by trying rule $\mathbf{r} \ 0 \ 1$.

Furthermore, *next_alt* accomplishes to another important task. Its behavior changes depending to the value of the boolean it receives in entry: *next_alt false A* aims to recursively cancel all the failures (if any) in A , whereas, *next_alt true A* aims to cancel the first success in A . This second behavior is helpful since, we need to progress the tree in case of success. For example, consider the tree in Figure 6f, it contains a success, mapping namely R to 2. The interpreter should return this success, but, due to the “non-deterministic” behavior of the prolog system, we also need to return all the other solution that could be available by evaluating the rest of the tree. After the replacement of the *path_end* in Figure 6f with *KO*, we remain with a tree having all failing path. The *next_alt* procedure keeps removing them until discovering that the tree have no worth-to explore paths, therefore it will return *None*.

Some interesting property of *next_alt* are shown below and allow to see how *next_alt* complements *step* wrt failed, success and *path_atom* trees.

$$\text{Lemma path_atom_next_alt_id: } \forall b \ A, \text{ path_atom } A \rightarrow \text{next_alt } b \ A = \bullet A. \quad (12)$$

$$\text{Lemma next_alt_failedF: } \forall b \ A \ A', \text{ next_alt } b \ A = \bullet A' \rightarrow \text{failed } A' = \perp. \quad (13)$$

$$\text{Lemma next_altFN_fail: } \forall A, \text{ next_alt } \perp \ A = \square \rightarrow \text{failed } A. \quad (14)$$

3.3 The *runT* inductive

The inductive *runT* is modeled as a function taking 4 inputs and returning 3 outputs. The inputs are the program, a set of variables v_0 , an initial substitution σ_0 , and a tree t_0 . The outputs are an optional object r containing a substitution σ_1 and an optional tree t_1 , and a boolean b and a set of updated variables v_1 . The very interesting piece of information is the result r , if *None*, it tells that the interpretation failed, otherwise, σ_1 represents the most-general unificator that makes the execution of the tree t_0 under the initial substitution σ_0 succeeds: σ_1 is an extension of σ_0 . The tree t_1 is the updated tree containing the alternatives that have not yet been explored, if these alternatives do not exists, t_1 is *None*. The output b and v_1 are there to tell respectively if during the execution a superficial cut has been evaluated, and what is the updated set of variables after the interpretation.

The procedure *runT* is based on four main derivation rules, shown in Figure 7. (STOPT) If the *path_end* of the tree t_0 is a success, the substitution σ_1 is retrived in t_0 from σ_0 and the input tree is cleaned of its successful path. (STEPT) If the *path_end* of t_0 is an atom, then *step* is invoked to evaluate t_0 , and *runT* is recursively called on the new tree. (BACKT)

248 If the *path_end* of t_0 is a failure, *next_alt* is called to clear the failed path; if the resulting
 249 cleaned tree t_1 exists, *runT* is recursively called on it. Finally, (FAILT) takes into account
 250 tree with no solutions.

251 **Lemma** *runT_det*: $\forall p \ v_0 \ \sigma_0 \ t_0 \ r_1 \ r_2 \ b_1 \ b_2 \ v_1 \ v_2,$
 $\text{runT } p \ v_0 \ \sigma_0 \ t_0 \ r_1 \ b_1 \ v_1 \rightarrow \text{runT } p \ v_0 \ \sigma_0 \ t_0 \ r_2 \ b_2 \ v_2 \rightarrow r_2 = r_1 \wedge v_2 = v_1 \wedge b_2 = b_1.$ (15)

252 The lemma above tells that *runT* behaves deterministically. The proof is by induction
 253 on the first *runT* execution and the proof is rather immediate, since all rules are in exclusive
 254 due to the hypothesis on the *path_end*: the hypotheses *success* t_0 , *path_atom* t_0 and *failed*
 255 t_0 are exclusive. Moreover, by Definition 15, we have that the hypothesis of FAILT implies
 256 *failed* t_0 , therefore it is exclusive wrt STOPT and STEPT, but also from BACKT since the
 257 have different expectations on the output of *next_alt*.

$$\begin{array}{c}
 \frac{\text{success } t_0 \quad \text{get_subst } \sigma_0 \ t_0 = \sigma_1 \quad \text{next_alt } \top \ t_0 = t_1}{\text{runT } v_0 \ \sigma_0 \ t_0 \ \bullet(\sigma_1, t_1) \perp v_0} \text{STOPT} \\
 \\
 \frac{\text{path_atom } t_0 \quad b_2 = \text{is_cb } \text{tg} \parallel b_1 \quad \text{step } p \ v_0 \ \sigma_0 \ t_0 = (v_1, \text{tg}, t_1) \quad \text{runT } v_1 \ \sigma_0 \ t_1 \ r \ b_1 \ v_2}{\text{runT } v_0 \ \sigma_0 \ t_0 \ r \ b_2 \ v_2} \text{STEPT} \\
 \\
 \frac{\text{failed } t_0 \quad \text{next_alt } \perp \ t_0 = \bullet t_1 \quad \text{runT } v_0 \ \sigma_0 \ t_1 \ r \ n \ v_1}{\text{runT } v_0 \ \sigma_0 \ t_0 \ r \ n \ v_1} \text{BACKT} \\
 \\
 \frac{\text{next_alt } \perp \ t_0 = \square}{\text{runT } v_0 \ \sigma_0 \ t_0 \ \square \perp v_0} \text{FAILT}
 \end{array}$$

■ **Figure 7** Rule system for *runT*

258 3.4 Properties of *runT*

259 Working with *runT* allows to prove some interesting properties about the behavior of the
 260 cut operator wrt disjunction, i.e. the *Or* node. We have proven several properties that are
 261 available at [this link](#). Below we show two of these lemmas.

262 **Lemma** *runSST_or*: $\forall p \ v_0 \ v_1 \ \sigma_0 \ \sigma_1 \ t_0 \ t_0' \ \text{sm } t_1,$
 $\text{runT } p \ v_0 \ \sigma_0 \ t_0 \ \bullet(\sigma_1, \bullet t_0') \top v_1 \rightarrow$ (16)
 $\text{runT } p \ v_0 \ \sigma_0 \ (\text{Or } \bullet t_0 \ \text{sm } t_1) \ \bullet(\sigma_1, \bullet(\text{Or } \bullet t_0' \ \text{sm } \text{KO})) \perp v_1.$

263 **Lemma** *runNT_or*: $\forall p \ v_0 \ v_1 \ \sigma_0 \ t_0 \ \text{sm } t_1,$
 $\text{runT } p \ v_0 \ \sigma_0 \ t_0 \ \square \top v_1 \rightarrow$ (17)
 $\text{runT } p \ v_0 \ \sigma_0 \ (\text{Or } \bullet t_0 \ \text{sm } t_1) \ \square \perp v_1.$

264 **Lemma** *run_orSST* $p \ v_0 \ v_2 \ \sigma_0 \ \sigma_1 \ \text{sm } t_0 \ t_0' \ t_1 \ t_1'$:
 $\text{runT } p \ v_0 \ \sigma_0 \ (\text{Or } \bullet t_0 \ \text{sm } t_1) \ \bullet(\sigma_1, \bullet(\text{Or } \bullet t_0' \ \text{sm } t_1')) \perp v_2 \rightarrow$ (18)
 $\exists b, \text{runT } p \ v_0 \ \sigma_0 \ t_0 \ \bullet(\sigma_1, \bullet t_0') \ b \ v_2 \wedge t_1' = \text{if } b \text{ then } \text{KO} \text{ else } t_1.$

265 In Definition 16, we evaluate the tree t_0 . This evaluation produces a success with signature
 266 σ_1 and a new tree t_0' . During the evaluation, a superficial cut in t_1 is traversed. This implies
 267 that the evaluation of $\bullet t_0 \vee_{s_m} t_1$ returns σ_1 as substitution, and the alternatives are $\bullet t_0' \vee_{s_m} \text{KO}$.
 268 The right-hand side of the *Or* node is replaced with *KO* by $!_r$.

269 In Definition 17, we are in a similar scenario: t_0 evaluates a superficial cut but, this time,
 270 produces a failure. This implies that the evaluation of $\bullet t_0 \vee_{s_m} t_1$ also fails.

Definition $\text{stepE } v_0 \text{ t s a gl} :=$
 $\text{let } (v_1, \text{rs}) := \text{bc p } v_0 \text{ t s in}$
 $(v_1, \text{save_alts a gl rs}).$

$$\frac{}{\text{runE } v_0 ((\sigma_0, \emptyset) :: a) \bullet(\sigma_0, a)} \text{STOPE}$$

$$\frac{\text{runE } v_0 ((\sigma_0, g) :: \text{ca}) \text{ r}}{\text{runE } v_0 ((\sigma_0, (\text{cut}, \text{ca}) :: g) :: _) \text{ r}} \text{CUTE}$$

$$\frac{\text{stepE } v_0 \text{ t } \sigma_0 \text{ a g} = (v_1, \text{bs}) \quad \text{runE } v_1 (\text{bs} ++ a) \text{ r}}{\text{runE } v_0 ((\sigma_0, (\text{call } t, _) :: g) :: a) \text{ r}} \text{CALLE}$$

$$\frac{}{\text{runE } _ \emptyset \square} \text{FAILE}$$

■ **Figure 8** Rule system for runE

Finally, in Definition 18, we evaluate $\bullet t_0 \vee_{s_m} t_1$. The evaluation succeeds, producing the result $\bullet(s_1, \bullet t'_0 \vee_{s_m} t'_1)$. This means that t_0 still has t'_0 as unexplored alternatives. We deduce that the evaluation of t_0 produces the substitution σ_1 with alternatives t'_0 . However, we have no information about whether a superficial cut has been crossed in t_0 . If this is the case, we know that t'_1 is cut away by $!_r$; otherwise, t'_1 has not been explored and is equal to t_1 .

From these lemmas, it is clear how the order in which alternatives are evaluated changes the behavior of the program.

4 Stack semantics

We now introduce the ELPI semantics. The interpreter is close to that of the ELPI language and provides an operational semantics that is similar to the one proposed by Pusch in [16], which is in turn closely related to the semantics given by Debray and Mishra in [6, Section 4.3]. Pusch mechanized this semantics in Isabelle/HOL, together with several optimizations that are also present in the Warren Abstract Machine [20, 1].

We represent a state of the ELPI language thanks to the two mutual inductives shown below.

Inductive $\text{alts} := \text{nilA} \mid \text{consA of } (\Sigma * \text{goals}) \ \& \ \text{alts}$
with $\text{goals} := \text{nilG} \mid \text{consG of } (A * \text{alts}) \ \& \ \text{goals} .$

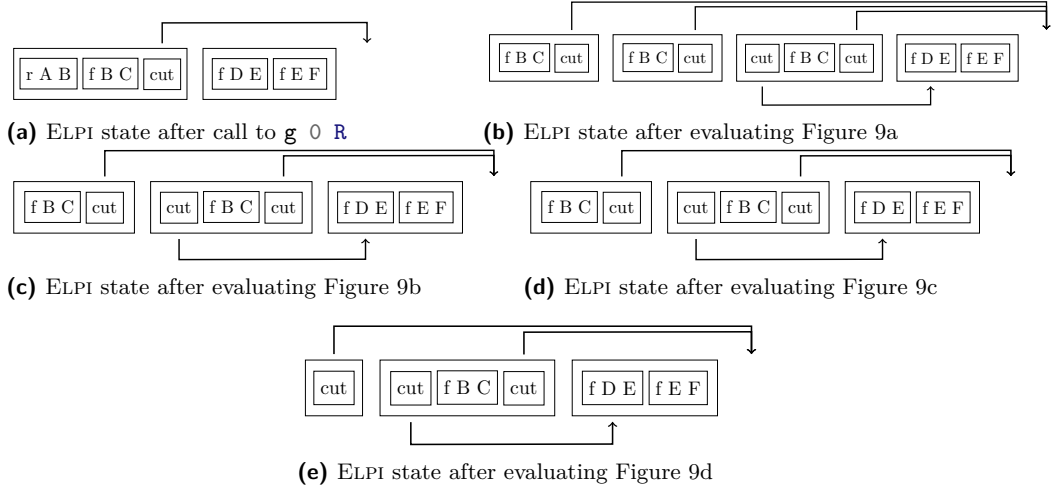
An ELPI state is an enhanced two-dimensional list. The outermost list represents a list of alternatives in disjunction, each accompanied by the substitution that should be used for their interpretation. The innermost list is a list of atoms, that is, the list of goals in conjunction. These goals are decorated with a pointer to an alternative list, which is used to keep track of where to jump when a cut is interpreted. We call these special alternatives the *cut-to* alternatives.

The ELPI interpreter is called runE and has the following type.

Inductive $\text{runE} : \mathcal{F}_v \rightarrow \text{alts} \rightarrow \text{option } (\Sigma * \text{alts}) \rightarrow \text{Prop} :=$ (19)

The inductive runE represents a routine taking as input a set of variables and a list of alternatives and returning an option. $\square$ stands for interpretation with no solution,

se riesco success
 $(A \text{ or } B) = \text{success } (B \text{ or } A)$ if
no sup cut in A
and B



■ **Figure 9** Example of ELPI execution

whereas $\bullet(\sigma, a)$ represent a successful interpretation with σ as solution and a as the list of non-yet-explored choice points.

The rule system for $runE$ is given in Figure 8 and is made by 4 cases. We distinguish two main cases: if the list of alternatives is empty, (FAILE) we have a failure: we stop the computation and return $\square$. If the list of alternatives is the list $(\sigma_0, h) :: a$, the interpreter evaluates the first choice point h under the substitution σ_0 . If h is empty, (STOPE), then we have a success, i.e. we return the input σ_0 and leave a as the future choice points. Otherwise, h is the list $(t, ca) :: g$ where t is the first atom to evaluate, ca is its cut-alternative list and g is the list of future conjuncts. If t is a cut (CUTE), we need to keep evaluating g under σ_0 but we change the alternatives to ca . Finally, in the case of a call (CALLE), we backchain the call in the program. If rs is the list of rule premises returned by bc , then each list of premises if mapped to a list of goals (goals) have a as cut-alternative and the global list is translated to a list of alternative (alts).

Example of execution We propose an exemple of execution of an ELPI state in Figure 9. The arrows in the images represent the cut-alternative pointer of the cut. We have not put arrow going our from atom-calls since the interpreter ignores the cut-alternatives in this case (cf. CALLE).

Remark We want finally to point out how the lemma that we were able to prove in Section 3.4 are not easy to be expressed in the ELPI semantics. This is beacuse there is no sufficient structure in the ELPI state to understand what is the scope of the cut operator. For example, in an ELPI state made of these alternatives $a_0, a_1 \dots a_n$, we have no clear idea of what happens if the first alternative a_0 succeeds: there could be a cut in a_0 but, is this cut superficial? What is its impact wrt the alternatives $a_1 \dots a_n$? More then that, it is also possible that the state is hill-formed and the cut does not point to a suffix in the list of alternatives.

5 Equivalence of the two semantics

In the previous section we have introduces the two semantics, they are The equivalence between the two semantics is possible under two conditions: we need to work with “valid tree”, i.e. all of those trees that can be generated from a call. Secondly, we need to translate

```

325 Fixpoint valid_tree A :=
326   match A with
327   | TA _ | OK | KO  $\Rightarrow$   $\top$ 
328   | Or  $\square$  _ B  $\Rightarrow$  valid_tree B
329   | Or  $\bullet$ A _ B  $\Rightarrow$  valid_tree A &&
330     ((B == KO) || B.base_or B)
331   | And A B0 B  $\Rightarrow$  valid_tree A &&
332     if success A then valid_tree B
333     else B == big_and B0
334   end.
335
336 Fixpoint t2l t0  $\sigma_0$  (cp: alts) : alts :=
337   match t0 with
338   | OK  $\Rightarrow$   $[(\sigma_0, \emptyset)]$ 
339   | KO  $\Rightarrow$   $\emptyset$ 
340   | TA a  $\Rightarrow$   $[(\sigma_0, [(a, \emptyset)])]$ 
341   ...

```

(a) Valid tree

(b) Tree to list

■ **Figure 10** Valid tree and Tree to list

325 trees into elpi states. In the next to subsection we propose the two functions *valid_state* and
 326 *t2l*.

327 5.1 Valid trees (*valid_state*)

328 The inductive tree allows one to generate a large number of trees, some of which are not
 329 valid, in the sense that they cannot be produced starting from a given query. The class of
 330 valid trees is characterized by the function shown in Figure 10a.

331 While all base cases of tree are considered valid, we need to analyze carefully the cases
 332 for the *Or* and *And* constructors.

333 For the *Or* constructor, we distinguish two cases depending on whether the lhs exists. If
 334 it does not exist, then the rhs must be a valid tree. Otherwise, the lhs must itself be a valid
 335 tree, and the rhs is either the *KO* tree, since it may have been removed by the evaluation of
 336 a superficial cut in the lhs, or it has not yet been explored. In the latter case, it is a *base_or*
 337 tree, namely the right-skewed tree formed by a disjunction of conjunctions that is generated
 338 after a successful backchain to a call.

339 For the *And* constructor, the lhs is required to be a valid tree. The shape of the rhs
 340 depends on whether the lhs is a success. If the lhs is not successful, then the rhs has never
 341 been explored: the procedures *step* and *next_alt* modify the rhs only when the lhs succeeds.
 342 In this case, the lhs must be the right-skewed tree containing conjunctions of premises atoms,
 343 the reset point B_0 ensure what shape the rhs should have. If the lhs is a success tree, then
 344 the rhs can be modified by *step* and *next_alt*, therefore it must be a valid tree.

345 5.2 From trees to lists (*t2l*)

346 The translation of a tree to a en ELPI state is done thanks to the *t2l* function. In Figure 10b
 347 we give just the implementation of the base cases, the full implementation is available at
 348 [this link](#).

349 The function *t2l* takes the tree t_0 we want to translate, a substitution σ_0 and a list of
 350 choice points *cp*. These choice points represent alternatives that appears at the same level
 351 of the current node, such as, for example, the right-siblings of a *Or* node. The *OK* node
 352 represent one successful alternative, therefore it is translated into a singleton list with the
 353 empty list of goals. The *KO* node represent a failure, i.e. it is mapped to the empty list of
 354 alternatives. Finally, atoms are mapped to one alternative containing a single goal with the
 355 empty cut-alternative list. They cannot point to *cp* since they are not right-uncle subtrees.

■ **Table 1** Tree to list from Figure 6 and Figure 9

t	Figure 6a	Figure 6b	Figures 6c and 6d	Figure 6e	Figure 6f
$t2l \ t \in \emptyset$	Figure 9a	Figure 9b	Figure 9c	Figure 9d	Figure 9e

We have omitted the code for the translation of the *Or* and *And* constructor since it is quite complex, but, we give the intuition behind this translation by means of examples.

The translation of $A \vee B_\sigma$ creates two list of alternatives, one for the A and the other for B . The alternatives of the rhs are valid choice points for the lhs and should be passed to the translation of A . The two list of alternatives can be concatenated and, each atom, should add cp as new cut-to alternatives: A and B are one level lower wrt cp and therefore the cut they have should point to cp .

The translation of $A \wedge_{B_0} B$ implies the translation of A . If the translation of A returns an empty list we return the empty list of alternatives without even touching B nor B_0 . Otherwise the translation returns a list $(\sigma_0, g) :: a$. By the semantics of *runT* the B node represent the continuation of the first alternatives in A , i.e. g , whereas B_0 is the continuation of goals for each alternative in a . While performing these translation, we need to carefully update the cut-alternatives.

Example of t2l In Table 1 we point out the result of calling *t2l* on the trees in Figure 6. The first row contains the image reference of then trees and the second row contains the reference of the images from the ELPI state in Figure 9. For the 3rd column in the table, we see that *t2l* is not injective. There are different trees that maps to the same ELPI state. In fact, there are transitions in *runT* that are not visible in *runE*.

5.3 Equivalence theorems

Thanks to the two functions *valid_state* and *t2l*, we are able to finally state the equivalence theorems between the semantics. The proof we are giving are on a slightly different definition of *runT*, since in the equivalence, we ignore the verification for the boolean and the variable set returned by *runT*, these piece of data are particularly meant for the proof in Section 3.4.

The definition we use is the following:

Definition *runT'* $p \ v \ s \ t \ r := \exists \ b \ f v, \text{runT } p \ v \ s \ t \ r \ b \ f v.$ (20)

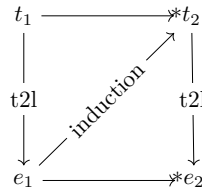
Before going to the equivalence proofs, we want to claim a key lemma allowing to ignore the silent transition in the *runT* inductive.

Lemma *next_altF_t2l*: $\forall \ t_0 \ t_1 \ b_0,$
 $\text{valid_tree } t_0 \rightarrow \text{failed } t_0 \rightarrow \text{next_alt } b_0 \ t_0 = \bullet t_1 \rightarrow$
 $\forall \ l \ \sigma_3, \ t2l \ t_0 \ \sigma_3 \ l = t2l \ t_1 \ \sigma_3 \ l.$ (21)

It is proved easily by induction on the tree t_0 .

Lemma *tree_to_elpi*: $\forall \ p \ t_0 \ \sigma_0 \ r,$
 $\text{let } v_0 := \text{vars_tree } t_0 \ \backslash \ \text{vars_sigma } \sigma_0 \text{ in}$
 $\text{valid_tree } t_0 \rightarrow$
 $\text{runT}' \ p \ v_0 \ \sigma_0 \ t_0 \ r \rightarrow$
 $\text{let } a := t2l \ t_0 \ \sigma_0 \ \emptyset \text{ in}$
 $\text{let } r := \text{if } r \text{ is } \bullet(s, t) \text{ then } \bullet(s, t2l \ (\text{odflt } K0 \ t) \ \sigma_0 \ \emptyset)$
 $\text{else } \square \text{ in}$
 $\text{runE } p \ v_0 \ a \ r.$ (22)

Proof sketch We proceed by induction on the execution of *runT'*. There are 4 cases to be treated. The first case comes from *STOPT*, it deals with successful trees, a successful tree must to start with an empty list and therefore we conclude with *STOPE*. The case for



■ **Figure 11** Induction scheme for Definition 23

389 STEPT, need to treat two cases: *step* has evaluated either a cut or a call. Depending on this
 390 we should use CUTE or CALLE and then the induction hypothesis. The case for BACKT is a
 391 silent case: it represents the non-injective behavior of *t2l*, but, thanks to Definition 21 and
 392 the induction hypotheses the conclusion is proved. The last case comes from the derivation
 393 FAILT. It is easily proved using FAILE and the fact that if *next_alt* returns $\square$ when trying
 394 to ease failure in t_0 , then *t2l* of t_0 returns the empty list.

Lemma *elpi_to_tree* $p \ v_0 \ a \ r :$
 $\text{runE } p \ v_0 \ a \ r \rightarrow$
 $\forall \sigma_0 \ t_0, \text{valid_tree } t_0 \rightarrow \text{t2l } t_0 \ \sigma_0 \ \emptyset = a \rightarrow$
 395 **if** r **is** $\bullet(\sigma_1, a')$ **then** (23)
 $\exists t_1, \text{runT}' \ p \ v_0 \ \sigma_0 \ t_0 \bullet(\sigma_1, t_1) \wedge \text{t2l } (\text{odflt } K0 \ t_1) \ \sigma_0 \ \emptyset = a'$
else $\text{runT}' \ p \ v_0 \ \sigma_0 \ t_0 \ \square$.

396 The proof of Definition 23 is based on the idea explained in [2, Section 3.3]. An ideal
 397 statement for this lemma would be: given a function *l2t* transforming an elpi state to a
 398 tree, we would have that the execution of an elpi state e is the same as executing
 399 *runT* on the tree resulting from *l2t*(e). However, it is difficult to retrieve the structure of
 400 an elpi state and create a tree from it. This is because, in an elpi state, we have no clear
 401 information about the scope of an atom inside the list and, therefore, no evident clue about
 402 where this atom should be placed in the tree.

403 Our theorem states that, starting from a valid tree t which translates to a list of alternatives
 404 $(\sigma_1, g) :: a$. If we run in elpi the list of alternatives, then the execution of the tree t returns the
 405 same result as the execution in elpi. The proof is performed by induction on the derivations
 406 of the elpi execution. We have 4 derivations.

407 We have 4 cases to analyse:

408 6 Case study: determinacy analysis

409 we mechanize the first order part of xxx.

410 snippet det, main thm, invariant det tree (valid tree prev section?)

411 proof induction on exec, step/next alt preserving invariant proved by induction on the
 412 tree.

413 with list semantics cut and next alt requires to express a link between the call or next alts
 414 and the current goal, which is non trivial without an intermediate data structure like the tree

415 7 Related work

416 The only mechanization of Prolog's semantics we could find is the one by Pusch in ISA-
 417 BELLE/HOL [16] that is based on the model of Debray and Mishra [6, Sec. 4.3]. Their work
 418 starts from that semantics and refines it by introducing some optimizations usually found

in the Warren abstract machine [20, 1]. They do not use the semantics to prove properties on the execution of programs as we do in our use case, hence they do not face the difficulty described in section XXX that pushed us to develop an more explicit semantics (with respect to cut). In some way our work is complementary, showing he stack based semantics equivalent to more optimized one, we could consider implement in Elpi.

Another relevant work are the ones by King cite, but we could not find their code. Their semantics is denotational and cannot easily cover all the features our paper XX does, but would have been sufficient for this first step.

The proof technique we use to relate the two semantics is inspired by XXX and we thank Y.Bertot for insightful discussions on the topic. In XXX Bertot relates bla bla, without never constructing a decompiler, exactly as we never build a list to tree function.

8 Conclusion

We do this. This is a first step for XXXX. We need to add substitutions. The missing part of the proof becomes related to abstract interpretation, since the det types for a lattice and one has to prove that the concrete runtime values variables take agree with the abstraction computed statically. This proof is ongoing and somehow orthogonal to the current one since the cut operator and the HO feature do not interfere with eachother in complex ways.

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